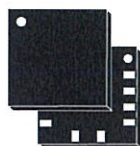


* 10/19/21

Digital output magnetic sensor: ultra-low-power, high-performance 3-axis magnetometer

Datasheet - production data



LGA-12 (2.0x2.0x0.7 mm)

Features

- 3 magnetic field channels
- ± 50 gauss magnetic dynamic range
- 16-bit data output
- SPI/I²C serial interfaces
- Analog supply voltage 1.71 V to 3.6 V
- Selectable power mode/resolution
- Single measurement mode
- Programmable interrupt generator
- Embedded self-test
- Embedded temperature sensor
- ECOPACK[®], RoHS and "Green" compliant

Applications

- Tilt-compensated compasses
- Map rotation
- Intelligent power saving for handheld devices
- Gaming and virtual reality input devices

Description

The LIS2MDL is an ultra-low-power, high-performance 3-axis digital magnetic sensor.

The LIS2MDL has a magnetic field dynamic range of ± 50 gauss.

The LIS2MDL includes an I²C serial bus interface that supports standard, fast mode, fast mode plus, and high-speed (100 kHz, 400 kHz, 1 MHz, and 3.4 MHz) and an SPI serial standard interface.

The device can be configured to generate an interrupt signal for magnetic field detection.

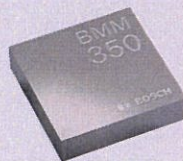
The LIS2MDL is available in a plastic land grid array package (LGA) and is guaranteed to operate over an extended temperature range from -40 °C to +85 °C.

Table 1. Device summary

Part number	Temp. range [°C]	Package	Packaging
LIS2MDL	-40 to +85	LGA-12	Tray
LIS2MDLTR	-40 to +85	LGA-12	Tape and reel

BMM350

3-axis magnetic sensor with high data rate



BST-BMM350-DS001-26

Document revision	1.26
Document release date	November 2024
Document number	BST-BMM350-DS001-26
Sales Part Number	0273.017.027
Notes	Data and descriptions in this document are subject to change without notice. Product photos and pictures are for illustration purposes only and may differ from the real product appearance.

1. Basic Description

The BMM350 is a 3-axis magnetic sensor which operates in automatic mode or triggered mode. The magnetic-to-digital conversion technology is based on TMR (tunnel magneto resistance)

The BMM350 is designed to meet all requirements for high performance consumer applications such as magnetic compass, virtual, augmented and mixed reality applications and high-end gaming, platform stabilization applications such as image stabilization, or indoor navigation and dead-reckoning, for example in robotics applications.

Evaluation circuitry converts the output of the magnetic transducer sensing structures, developed, produced and tested in BOSCH facilities. The corresponding chip-sets are packed into one single WLCSP 1.28mm x 1.28mm x 0.5mm housing. For optimum system integration the BMM350 is fitted with digital interfaces (I2C, I3C), offering a wide VDDIO voltage range from 1.72V to 3.6V. To provide maximum performance and reliability each device is tested and ready-to-use calibrated.

The BMM350 has an excellent temperature behaviour with an outstanding low temperature coefficients of the offset (TCO) and of the sensitivity (TCS).

An API is provided for quick integration into the target system.

Key features

Three-axis magnetic field sensor

- | | |
|--------------------------------|--|
| • Ultra-Small package | Wafer Level Chip Scale Package
(9 pins, 0.4mm ball pitch)
footprint 1.28 x 1.28 mm ² , height 0.5 mm |
| • Digital interface | I2C, I3C, 1 interrupt pin, 1 pin for I2C/I3C (legacy) address definition |
| • Low voltage operation | V _{DD} supply voltage range: 1.72V to 1.98V
V _{DDIO} interface voltage range: 1.72V to 3.6V |
| • Flexible functionality | Magnetic field range typical:
±2000µT (x, y-axis), ±2000µT (z-axis)
Magnetic field resolution of ~0.1µT |
| • On-chip interrupt controller | Interrupt-signal generation for new data |
| • Ultra-low power | Low current consumption (typ. 200µA @ 100 Hz in regular power preset), short wake-up time |
| • Normal and forced mode | periodic operation (normal mode) and triggered operation (forced mode) |
| • Temperature range | -40 °C ... +85 °C |
| • RoHS compliant, halogen-free | |